

PHISHING EMAIL DETECTION & ANALYSIS

PROJECT 1 — ACCOUNT VERIFICATION PHISHING EMAIL

1. Project Overview

This project demonstrates a practical SOC investigation of a controlled phishing-style email. The investigation focuses on email header analysis, authentication validation, IOC extraction, threat intelligence, social-engineering analysis, and evidence-based assessment.

2. Objective

- Analyze suspicious email headers and content.
- Validate SPF, DKIM, and DMARC.
- Extract relevant IOCs.
- Investigate the observed IP address.
- Identify phishing and social-engineering indicators.
- Correlate technical and behavioral evidence.
- Determine an evidence-based final assessment.

3. Email Details

From: luke <phishinglab1@gmail.com>

To: phishinglabreceiver@gmail.com

Subject: URGENT: Account Verification Required

Reply-To: luke <phishinglab1@gmail.com>

User-Agent: eMClient/10.4.0.0

4. Email Authentication Analysis

SPF: PASS

DKIM: PASS

DMARC: PASS

DARA: PASS

The authentication results indicate that the email passed the available SPF, DKIM, and DMARC checks.

The **From**, **Reply-To**, and **Return-Path** addresses are also consistent in this controlled test.

5. Source IP Analysis

Observed IP: 209.85.220.41

The IP was extracted directly from the email's Received and Received-SPF headers.

Hostname: mail-sor-f41.google.com

The IP is associated with **Google LLC** and **AS15169**.

6. IOC Extraction

Email Address: phishinglab1@gmail.com

IP Address: 209.85.220.41

Domain: gmail.com

URL: https://example.com/account-verification

7. Threat Intelligence Analysis

The IP address was investigated using threat-intelligence information.

ISP / Organization: Google LLC

ASN: AS15169

Hostname: mail-sor-f41.google.com

Abuse Reports: 514

Confidence of Abuse: 0%

The IP belongs to Google infrastructure. Although historical abuse reports are present, the current confidence-of-abuse value is 0%. Therefore, the IP alone is **not sufficient evidence to classify the email as malicious**.

8. Social-Engineering Analysis

The email contains several phishing-style characteristics:

- **Urgency** — "URGENT" is used in the subject.
- **Account-security pretext** — unusual activity is claimed.
- **Immediate action request** — the recipient is asked to verify the account.
- **Account suspension threat** — restricted access is mentioned.
- **Verification link** — a URL is included.

9. URL Analysis

URL: https://example.com/account-verification

The URL uses **example.com**, which is a reserved documentation/test domain.

Therefore, the URL is treated as a **controlled laboratory indicator**, not confirmed malicious infrastructure.

10. Final Assessment

Classification: PHISHING-STYLE TEST EMAIL

Final Verdict: Not confirmed malicious based on the available evidence.

The email demonstrates multiple phishing-style social-engineering techniques, but the technical evidence does not confirm a real malicious campaign.

PROJECT 2 — PASSWORD RESET / SECURITY ALERT PHISHING EMAIL

1. Project Overview

The second project is a separate controlled phishing-style email designed to demonstrate detection of stronger social-engineering techniques and compare them with the first test case.

2. Objective

- Analyze another phishing-style email.
- Identify social-engineering techniques.
- Extract relevant indicators.
- Analyze the email's raw source.
- Validate authentication results from the actual headers.
- Investigate suspicious indicators.
- Compare the findings with Project 1.

3. Email Details

To: phishinglabreceiver@gmail.com

Subject: Security Alert: Password Reset Required

4. Email Content

The email claims that a security issue has been detected and asks the recipient to verify their password immediately.

It uses:

- **Security alert**
- **Immediate verification**
- **Account-lock threat**
- **24-hour deadline**
- **Restricted-access warning**
- **Verification URL**

5. URL

URL: <https://example.com/security-verification>

The URL uses the controlled example.com domain and is therefore not treated as confirmed malicious infrastructure.

6. Social-Engineering Analysis

The second email contains stronger pressure tactics than Project 1:

- **Urgency**
- **Password verification request**
- **Account-lock threat**

- **Time-based pressure**
- **Restricted-access warning**
- **Security-alert pretext**

These characteristics are commonly associated with phishing-style social engineering.

7. Technical Investigation

The raw source of Project 2 should be analyzed for:

From

Reply-To

Return-Path

Received headers

Source IP

SPF

DKIM

DMARC

URLs

Domains

Only values actually present in the raw source should be recorded.

8. IOC Analysis

The primary observable indicator in this controlled test is:

URL: <https://example.com/security-verification>

Additional indicators should be extracted from the actual raw email source rather than assumed.

9. Evidence Correlation

The investigation follows:

Email Content

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Social-Engineering Indicators

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Raw Header Analysis

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Authentication Analysis

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IOC Extraction



Threat Intelligence



Evidence Correlation



Final Assessment

10. Final Assessment

Classification: CONTROLLED PHISHING-STYLE TEST EMAIL

The email demonstrates multiple phishing characteristics, particularly urgency, password verification, account-lock threats, and time pressure.

However, because this is a controlled laboratory email using a test URL, it should **not be represented as a real malicious phishing campaign**.

PROJECT COMPARISON

Area	Project 1	Project 2
Primary theme	Account verification	Password/security alert
Urgency	Present	Strong
Verification request	Present	Present
Account threat	Suspension/restricted access	Account lock/restricted access
URL	Controlled test URL	Controlled test URL
Header analysis	Performed	To be performed from actual raw source
IOC extraction	Performed	Performed from actual source
Threat intelligence	IP investigated	Investigate actual extracted indicators
Overall purpose	Baseline phishing-style analysis	Comparative phishing-style analysis

OVERALL SOC INVESTIGATION WORKFLOW

Email Collection



Raw Email Analysis



Header Analysis



SPF / DKIM / DMARC Validation



IOC Extraction



Threat Intelligence Investigation



Social-Engineering Analysis



Evidence Correlation



Risk Assessment



Final Verdict